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WEB TELEMETRY DASHBOARD AND 2D INCIDENT SIMULATOR

Full Instrument Cluster, 30s Real-Time Strip Recorder, Kinematic Trajectory Simulator & Replayer



Figure 4.1: *SECONDS AHEAD Real Telemetry Instrument Cluster Dashboard with 30-Second Multi-Trace Strip Recorder and Raw Serial Stream.*

Instrument Cluster Features & A-to-Z Telemetry Breakdown

Callout Zone	Instrument Element	Operational Function & Verified Telemetry Response
`[A] Connection Status`	Web Serial Interface	Streams live ESP32-H2 USB CDC telemetry at 115200 baud, 8N1. Verified throughput 18.0 PPS, system uptime 00:39:33.9.
`[B] Threat Banner`	Threat Evaluation State	Real-time threat state engine dynamically classifies WAIT, NO APPROACH, UNCERTAIN, MONITOR, and HIGH RISK.
`[C] Readout Tiles`	Multi-Sector Gauges	Target Distance: 37.8 cm (Danger <50cm) | Speed: 0.0 cm/s (Static) | TTC: Inactive | Lateral Sector B: 30.0 cm (Hazard <50cm).
`[D] Proximity Scale`	Range Scale (0-100 cm)	Visual gauge highlighting 0-50 cm danger threshold with dynamic numeric position slider at 37.8 cm.
`[E] Strip Chart`	30s Real-Time Recorder	White curve = Front Dist (A), Cyan curve = Lateral Dist (B), Dashed Amber = Speed (cm/s), Dashed Red = 50cm Limit.
`[F] Raw Serial Stream`	Diagnostic Stream	Streams raw hardware boot logs and serial packets directly from microcontroller ROM: [00:00.00] ESP-ROM: esp32h2-20221101.

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2D KINEMATIC INCIDENT SIMULATOR AND SYNCHRONIZED SPLIT VIEW

Physics-Based Trajectory Reconstruction, Dynamic Radar Cones, Impact Shockwave & Replayer Controls



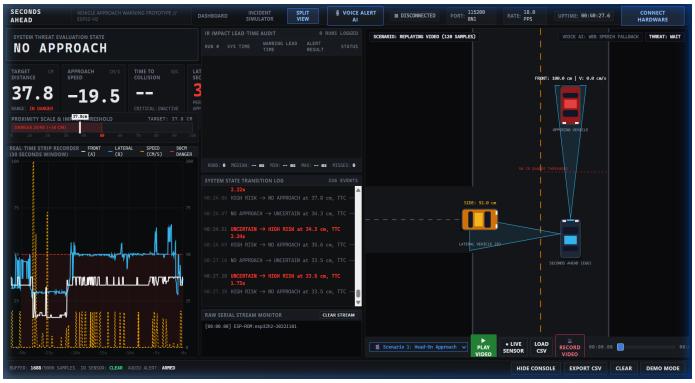


Figure 4.2 (Left): *2D Kinematic Incident Simulator View.* ***Figure 4.3 (Right):*** *Real-Time Synchronized Split View.*

2D Kinematic Simulation & Replayer Features

Callout Zone	Kinematic Feature	Mathematical Formulation & Dynamic Simulation Behavior
`[G] Ego Vehicle`	Test Vehicle Reference	Fixed reference coordinate at (egoX, egoY). Equipped with forward acoustic radar beam and side radar emitter.
`[H] Front Radar Cone`	Primary Sensor A Cone	Dynamic conical acoustic beam expanding Y_beam = dist * 2.8px. Color-coded Cyan (Nominal) -> Red (Threat).
`[I] Lateral Radar Cone`	Perpendicular Sensor B	90° side blind-spot beam monitoring perpendicular cross-street intersection traffic up to 100 cm.
`[J] Opposing Vehicle`	Longitudinal Approaching Car	Position computed via: Y_opp = Y_ego - (dist * 2.8px). Tagged with live approach speed (cm/s) and TTC.
`[K] Lateral Vehicle`	Cross-Traffic Obstacle	Position computed via: X_side = X_ego - (side_dist * 2.2px). Simulates perpendicular blind-spot hazard.
`[L] Impact Shockwave`	Ground-Truth Contact Burst	Expanding orange/red shockwave rings rendered instantaneously when TCRT5000 IR sensor registers contact.
`[M] Replayer Controls`	Incident Timeline Bar	Timeline scrubber slider, 0.5x / 1x / 2x playback rates, CSV log loader, and WebM video clip recorder.
`[N] Voice Alert AI`	ElevenLabs Neural Audio	Synthesizes urgent spoken warnings: 'Critical warning! Collision in 1.8 seconds! Brake immediately!'.